PRINCIPLES OF ECONOMICS

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FPEC 101 – LECTURE 7

Unit 2

PRINCIPLES OF ECONOMICS

Production Costs and Market Structures

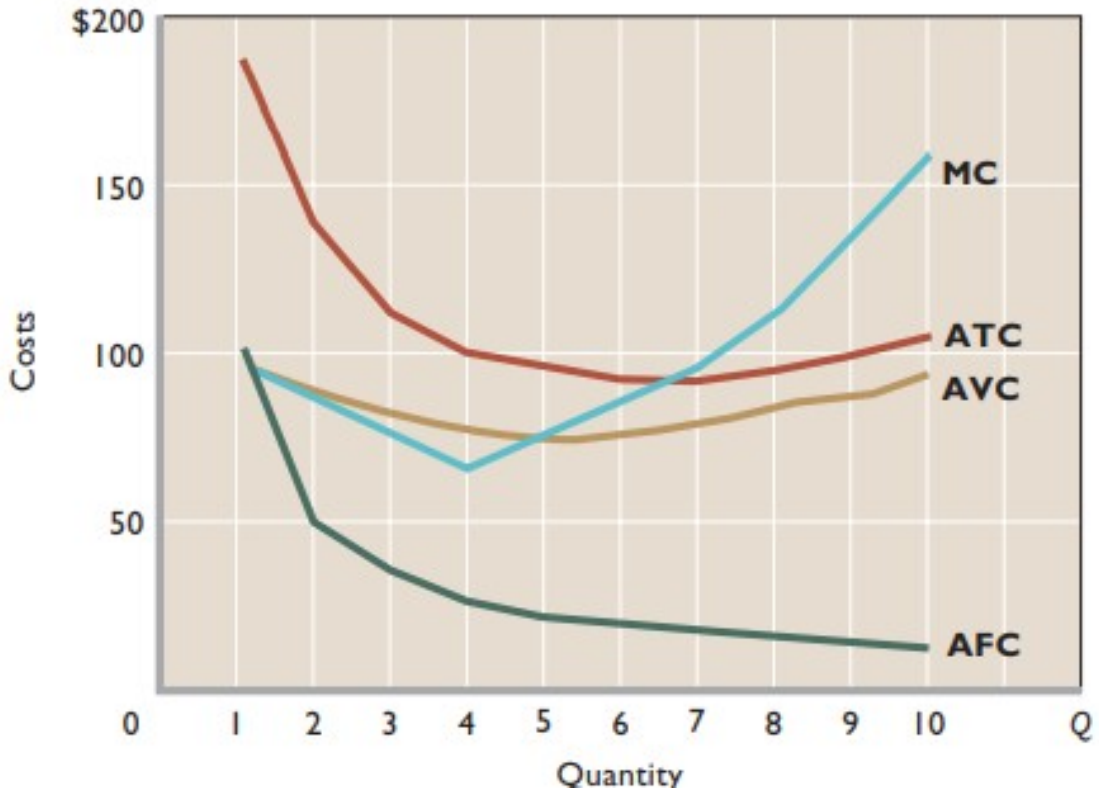
FPEC 101 – LECTURE 7

Unit 2

Lecture Outline
Economic, Short and Long Run costs

- Short Run and Long run Costs
- Legal Forms of Business
- Growth of Firms

Short Run Costs- Average Costs



- MC cuts ATC and AVC at minimum.
- When MC is below ATC, ATC falls
- When MC is above ATC, ATC rises.

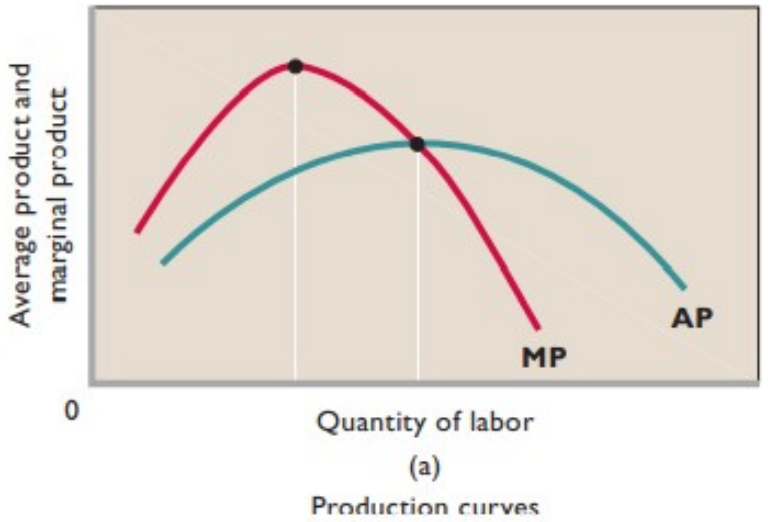
- Similarly, when MC is below AVC,AVC falls.
- When MC is above AVC, AVC rises.

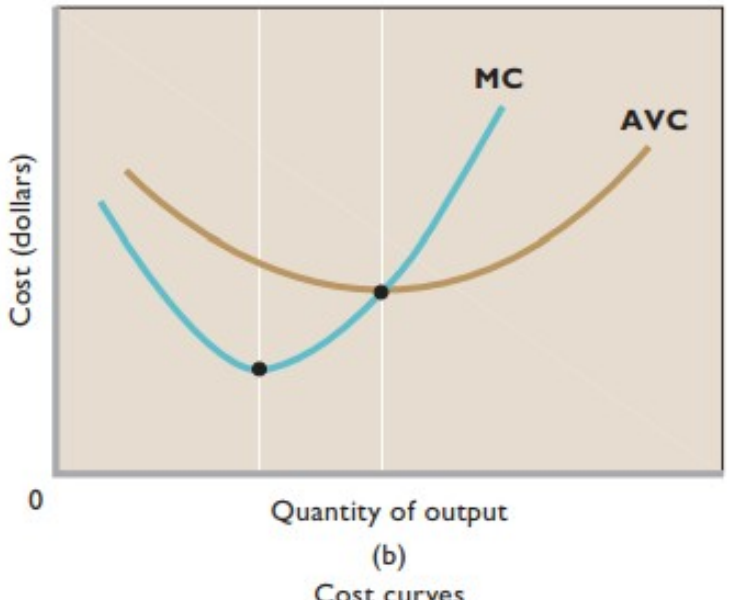
Mc Connell & Brue pg. 388

Short Run Costs-
Relation of MC to AVC and ATC

- MC intersects cuts the AVC and the ATC curves at their minimum points.
- When MC is added to total cost is less than the current average total cost, ATC will fall and vice versa.
- As long as MC lies below ATC, ATC will fall, and whenever MC lies above ATC, ATC will rise.
- Where MC equals ATC, ATC is constant and thus is the minimum point on the ATC curve.
- MC cuts ATC at ATC curve's minimum point
- With the same logic MC curve cuts AVC at AVCs minimum'.
- No such relationship exists between the MC curve and the average-fixed-cost curve, because the two are not related; marginal cost includes only those costs that change with output.

Relation between productivity and cost cu





- MC and AVC are mirror images of the MP and AP curves.
- Assuming labor as only variable input and constant wage rate. when MP is rising, MC is falling, and when MP is falling, MC is rising.
- Under the same assumptions, when AP is rising, AVC is falling, and when AP is falling, AVC is rising

Q	TFC	TVC	TC	AFC	AVC	ATC	MC
0	100	0	100				
1	100	90	190				
2	100	170	270				
3	100	240	340				
4	100	300	400				
5	100	370	470				
6	100	450	550				
7	100	540	640				
8	100	650	750				
9	100	780	880				
10	100	930	1030				

Q	TFC	TVC	TC	AFC	AVC	ATC	MC
0	100	0	100				
1	100	90	190	100	90	190	
2	100	170	270	50	85	135	
3	100	240	340	33.33	80	113.33	
4	100	300	400	25	75	100	
5	100	370	470	20	74	94	
6	100	450	550	16.67	75	91.67	
7	100	540	640	14.29	77.14	91.43	
8	100	650	750	12.50	81.25	93.75	
9	100	780	880	11.11	86.67	97.78	
10	100	930	1030	10	93	103	

AFC=TFC / Q

AVC=TVC / Q

ATC=TC / Q

Q	TFC	TVC	TC	AFC	AVC	ATC	MC
0	100	0					
1	100	90					90
2	100	170					80
3	100	240					70
4	100	300					60
5	100	370					70
6	100	450					80
7	100	540					90
8	100	650					110
9	100	780					130
10	100	930					150

2.5

MC is graphed at average Q

$MC = \Delta TC / \Delta Q$

Shifts of Cost Curves

Change in Factor Prices

If price of fixed input increases

- AFC and ATC shift up
- AVC and MC unchanged

If price of variable input increases

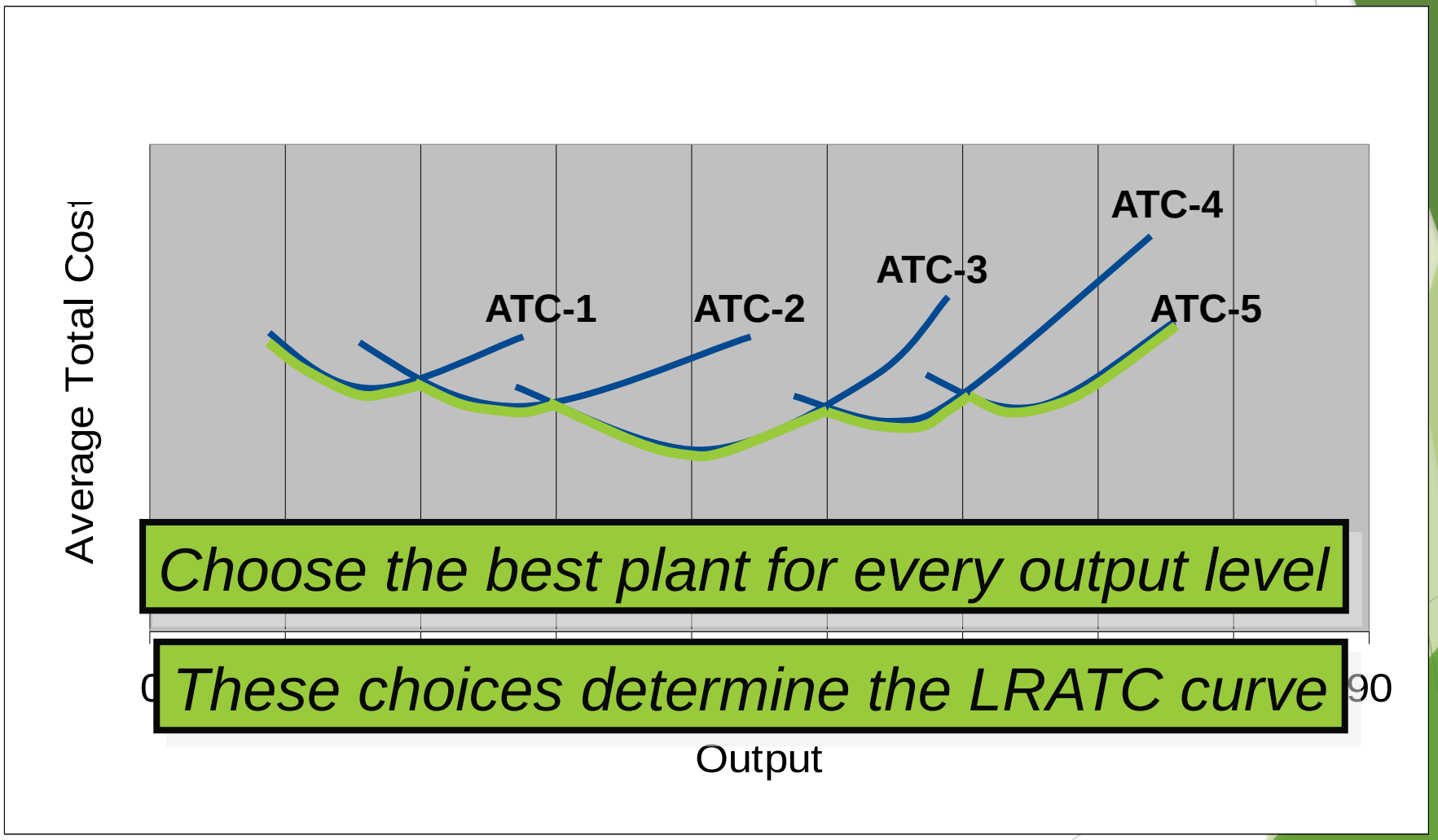
- AVC, ATC, and MC shift up
- AFC unchanged

- If change in technology, curve shifts depends on whether technology affects FC, VC, or both

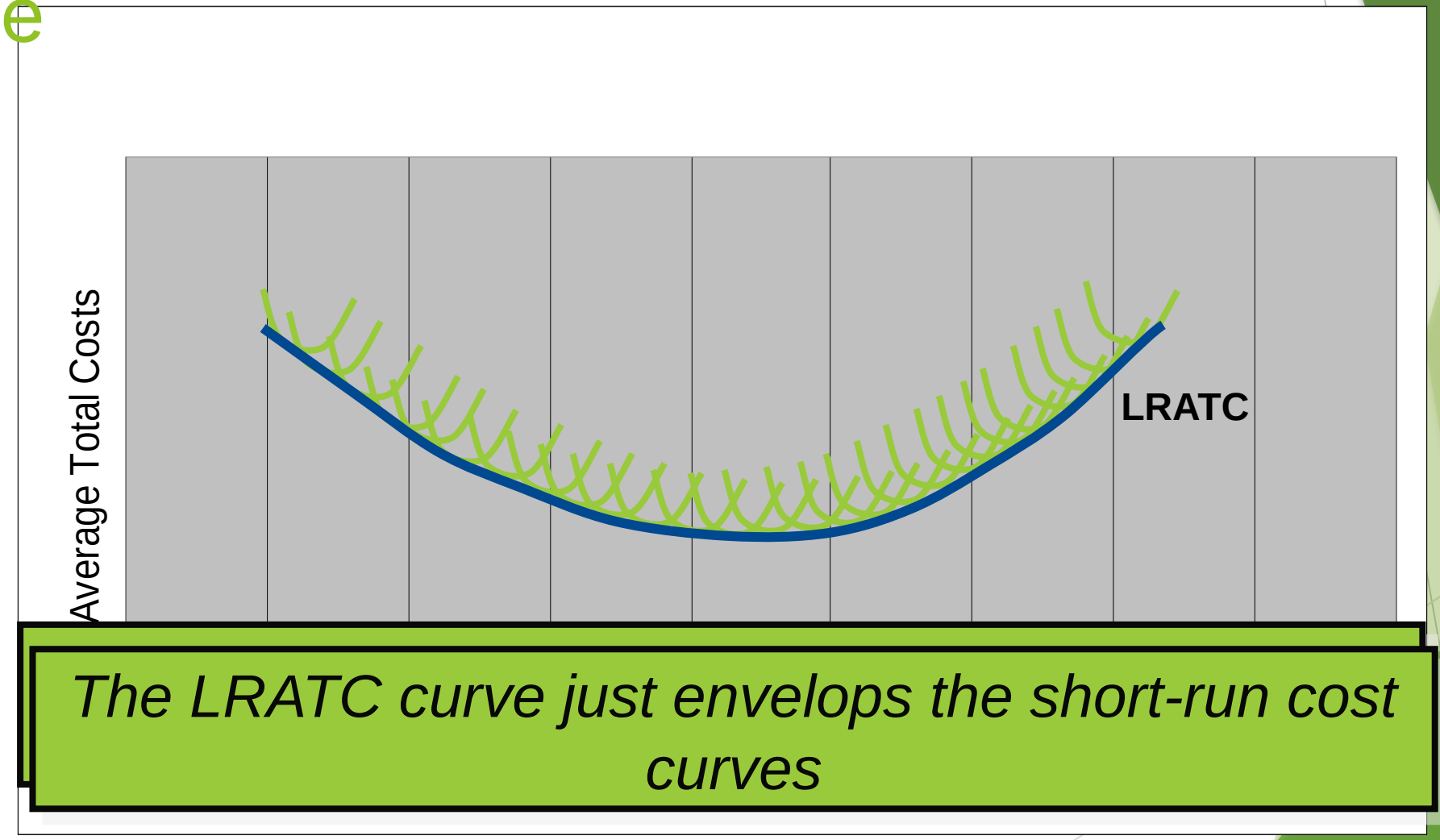
Long Run Production Costs

- Since it is the Long run, we focus on Average Total Cost because all resources, and, therefore, costs are variable.
- For a time, successively larger plants will lower average total cost. However, eventually the building of a still larger plant may cause ATC to rise.
- Initially a business has a small plant size up to a certain level of output. When ATC reaches minimum and begins rising, firm should change plant size. For the next level of output, the firm attains lower costs with a bigger plant size. This continues as the firm expands and eventually reaches a minimum to rise again.
- Tracing these adjustments, we find that the long-run ATC curve for the enterprise is made up of segments of the short-run ATC curves for the various plant sizes that can be constructed.

Long-Run Average-Total-Cost Curve



The Long-Run Average-Total-Cost Curve



The LRATC curve just envelops the short-run cost curves

The long-run ATC curve is made up of all the points of tangency of the unlimited number of short-run ATC curves

Long-Run Costs

- ***Economies*** of large scale refer to the advantages a firm accrues from producing on a large scale which results in a fall in the long run average total cost or per unit cost.
- ***Diseconomies of scale*** conversely refer to the rise in average costs as production is increased beyond the optimum scale.
- **Constant returns to scale** is the point when economies of scale end and the output at which diseconomies of scale begin. However, there may be a range of constant returns to scale over which long-run average cost does not change
- **Minimum efficient scale** (MES), which is the lowest level of output at which a firm can minimize long-run average costs.
- The best or optimum size for a firm would be where it can reduce the average cost to their lowest point in the long run. At this point it can benefit from economies of scale and use the profits to improve their factory and their products or to lower prices to attract consumers away from competing firms.

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Long-Run Costs

- ***Economies* of *scale*** .
  - Technical economies :
  - Financial economies
  - Marketing economies
  - Purchasing economies
  - Managerial economies
  - Risk bearing economies

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Long-Run Costs

- ***Diseconomies* of *scale*** .
  - Management problems
  - Labor alienation
  - Price of inputs
  - Lack of personal contact

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Growth of Firms

- Reasons
- Growth
  - Internal
  - External
    - Take over/ acquisition
    - Merger / Amalgamation
    - Conglomerate/ Lateral

Legal Forms of Business

- Sole Trader
- Partnership
- Joint stock companies- Public, private
- Corporations

SUMMARY

- Short Run Costs
  - Concept of Total, Average and Marginal Products
  - Concept of Fixed, Variable and Total Cost
  - Average Costs
- Relationship between curves
- Long run Costs
  - Economies and diseconomies of scale
- Legal forms of business
  - Sole Trader
  - Partnership
  - Joint stock companies- Public, private
  - Corporations
- Growth of firms